

Microfinance Loan Eligibility Criteria & Credit Risk Framework

1. Overview of Microfinance Credit Assessment

Microfinance institutions provide financial access to underserved individuals and micro-enterprises lacking formal collateral. Underwriting requires a blend of financial metrics and alternative credit data.

2. Core Eligibility & Risk Evaluation Metrics

Risk Dimension	Primary Indicator	Threshold Standard	Weight in Score
Repayment Capacity	Debt-to-Income Ratio (DTI)	<= 40% of net monthly income	35%
Credit Track Record	Past Repayment Rate & Delinquencies	0 defaults in last 12 months	25%
Business Stability	Operating History of Enterprise	>= 12 months active operations	20%
Alternative Data	Utility Payment History / Mobile Money	Consistent on-time payments	10%
Social Collateral	Joint Liability Group (JLG) Guarantee	Peer validation from 4+ members	10%

3. Decision Pipeline Architecture

- 1. Application Ingestion:** Collect applicant demographics, business revenue, and repayment history.
- 2. Rule-Based Pre-Screening:** Filter out underage applicants (< 18) or individuals with active major defaults.
- 3. ML Scoring Model:** Random Forest classifier predicts default probability (0.00 to 1.00).
- 4. GenAI Plain-English Explainer:** Translates model feature importances into transparent, respectful explanations for applicants.